

Methodology for Predicting Axial Load Capacity of CFST Columns

1. Preliminary Problem Formulation

The primary objective is to accurately predict the axial load capacity of Concrete-Filled Steel Tube (CFST) columns using advanced data-driven machine learning techniques. Traditional empirical structural design codes (e.g., IS 456, IS 800, Eurocode 4) frequently rely on simplified formulations that can yield overly conservative results or high scatter for extreme geometries. This research establishes a robust supervised learning framework to overcome these limitations.

2. Data Collection and Preprocessing

- Data Aggregation: An experimental database of CFST columns was compiled and structured (CIRC_model_ready.csv).
- Feature Extraction: The primary structural features isolated for modeling include Geometric Parameters (Column Length L , Tube Diameter D , Tube Thickness t) and Material Properties (Concrete Compressive Strength f_c , Steel Yield Strength f_y).
- Data Preparation: The input features and target variables were standardized using a scaler to ensure uniform gradient propagation. The dataset was split into training and testing sets.

3. Baseline Modeling (Data-Driven)

- Algorithm Selection: Two distinct supervised learning algorithms were deployed: CatBoost (a robust tree-based Gradient Boosting algorithm) and an Artificial Neural Network (ANN) constructed using PyTorch with ReLU activation and Dropout layers.
- Optimization: Network architectures and hyperparameters were tuned to minimize prediction error. The optimal model weights were saved (optimized_ann_model.pt and best_model.pt).

4. Advanced Analysis and Evaluation

- Statistical Performance Evaluation: Models were benchmarked using R-squared, RMSE, and MAE against true experimental capacities.
- Physical Interpretability (SHAP): SHAP values were applied to extract feature importance, validating that the models autonomously learned structural mechanics like volumetric scaling and slenderness penalties.
- Uncertainty Quantification (Monte Carlo Dropout): Epistemic uncertainty was quantified by executing Monte Carlo Dropout during inference, generating a predictive distribution across 100 stochastic forward passes.

5. Comparative Analysis and Generalization Check

- The predictive performance and error margins of the optimized ANN were benchmarked against the CatBoost model.
- Outliers with high uncertainty (identified via Monte Carlo evaluation) were analyzed to map the boundary limitations of purely data-driven structural models.